Following **T 1063/06**, the board held in **T 852/09** that since the enhancers to be used were characterised in functional terms only and the claim merely represented for the skilled person an invitation to perform a research programme, he/she could not carry out the invention within the entire scope claimed without undue burden (see also **T 155/08**).

In **T 2070/13** the board indicated in its preliminary opinion that an adequate test method for anti-adherence was lacking and that no suitable compounds were defined within the claimed families of compounds. In particular, the argument of the respondent (patent proprietor) that the skilled person could carry out the invention by way of routine optimisation using standard procedures was not convincing. The board stated that no standard procedure was disclosed in the patent in suit and none was referred to during the whole proceedings. Even if a specific model test procedure were used to determine the anti-adherence, this did not significantly reduce the extent of the experimental programme required to identify the appropriate anti-adherent materials. The board concluded that the skilled person was unable to identify suitable materials displaying anti-adherence due to the large number of potential materials listed, several of these listed materials also describing large families of compounds; the skilled person would be faced with an experimental programme in order to establish which of the listed materials satisfied the claimed criterion of anti-adherence.

If the patent claims require that a specific aim should be achieved (in this case, that a specific value of a parameter should not be reached), then there is no guarantee that the invention can be carried out in accordance with Art. 100(b) EPC if the patent affords the skilled person no clue as to how he can achieve this aim outside the scope of the embodiments without an undue burden of research (**T 809/07**).

In **T 518/17** the board stated that a reasonable amount of trial-and-error experimentation might be acceptable for the purposes of finding that a claimed invention could be carried out without undue burden. This, however, presupposed that sufficient information was available that led the skilled person directly towards success through the evaluation of initial failures. Based on the evidence on file, the board considered that information on critical process variables was lacking here. The skilled person would therefore need to repeat the lengthy optimisation and improvement process, without any guarantee of success. Such a situation was often described in the case law as an invitation to perform a "research program" and considered to amount to an undue burden.

6.8. Post-published documents

Sufficiency of disclosure must, in principle, exist at the effective date of a patent, while post-published documents may be used as evidence that the disclosure is reproducible without undue burden only under certain circumstances.

In the absence of any tangible proof in the patent specification that the claimed concept can be put into practice, post-published documents can be used as evidence whether the invention merely disclosed at a general conceptual level was indeed reproducible without undue burden at the relevant filing date (**T 994/95** and **T 157/03**). In **T 1262/04** of 7 March 2007, the board considered that this principle applied at least to cases such as the one at issue, where the technical teaching as disclosed in the application was credible. In **T 1205/07**, the post-published documents were considered, as the evidence they provided was not aimed at "curing" any alleged insufficiency of disclosure, but rather at confirming the teachings of the application. See also **T 1547/08**.

Even though sufficiency of disclosure must, in principle, be established at the priority date, post-published documents can be used as evidence that the claimed concept can be put in practice. Accordingly, the board decided to consider documents in spite of their late filing (**T 1164/11**).

The fact that experimental data were not published until after the filing date of the application is not prejudicial to their nature as evidence of physical phenomena, which occur independently of any publication date (**T 416/14**).

If a disclosure is seriously insufficient in that it provides no guidance for performing a particular aspect of the invention, a reference to later documents showing how such performance was accomplished at a later date is manifestly incapable of curing the insufficiency (**T 222/00**). Sufficiency of disclosure must, in principle, be shown to exist at the effective date of a patent. If the description of the patent specification provides no more than a vague indication of a possible medical use for a chemical compound yet to be identified, later more detailed evidence cannot be used to remedy the fundamental insufficiency of disclosure of such subject-matter (**T 609/02**). The disclosure in post-published documents can only be taken into account for the question of sufficiency of disclosure if it was used to back up the positive findings in relation to the disclosure in a patent application (**T 1273/09** citing **T 609/02**). Post-published evidence may be taken into account, but only to back-up the findings in the application in relation to the use of the compound(s) as a pharmaceutical (**T 609/02**, **T 950/13**, see in this chapter II.C.7.2.2c)).

In case **T 1329/11** the respondents (patent proprietors) referred to post-published documents, in particular to document D8 published more than five years after the priority date, in order to show that the claimed method worked. The contents of documents which were not available to the skilled person at the priority date could not help to overcome the major problem of sufficiency of disclosure of the claimed invention at the priority date.

In **T 2070/13** the board observed that D16 – a patent specification – failed to provide any guidance as to how anti-adherence might be determined; the document was post-published with respect to the patent in suit and its disclosure was thus of no relevance to the sufficiency of disclosure thereof.

In **T 1255/11** the board was satisfied that the application as filed provided a complete theoretical explanation, backed up by scientific literature, for the treatment of Alzheimer's disease by MCT. Since the presence of the claimed effect was made plausible by the theoretical background explanations provided in the application as filed, the appellant (patent proprietor) might provide post-published evidence.

T 59/18 addressed the issue of series of patent specifications as a means of establishing common general knowledge. The board ultimately decided that any possible definition of the term "relaxation ratio" found in post-published documents could not be regarded as an indirect indication of common general knowledge in the absence of a specific indication that the meaning of this term already belonged to older prior art. Only one of the patent documents submitted by the appellant contained a definition of a relaxation ratio. It could not be found on that basis that there were a series of patent specifications providing a consistent picture within the meaning of **T 412/09**.

In **T 116/18** (OJ 2022, A76) the issue of post-published evidence led to a referral to the Enlarged Board (**G 2/21**). In essence, this referral was concerned with inventive step (Art. 56 EPC) – the patent proprietor had cited a piece of post-published evidence to demonstrate that the problem was solved and that the alleged technical effect was achieved. The referral contains a definition by the board of the term "post-published evidence". The board identified three main lines of case law: **T 1329/04**, **T 609/02**, **T 488/16**, **T 415/11**, **T 1791/11** and **T 895/13** of 21 May 2015 ("ab initio plausibility" – plausibility was ultimately not established in these decisions); **T 919/15**, **T 578/06**, **T 2015/20**, **T 536/07**, **T 1437/07**, **T 266/10**, **T 863/12**, **T 184/16** ("ab initio implausibility" – plausibility was ultimately established in these decisions); **T 31/18**, **T 2371/13** ("no plausibility" line of case law). It also stated that the issue whether post-published evidence could be taken into account arose under the heading of sufficiency of disclosure too, where the effect was expressed in the claim at issue. Whether an effect was part of the problem to be solved or was expressed in the claim at issue dictated which provision of the EPC was applicable (**G 1/03**, OJ 2004, 413, point 2.5.2 of the Reasons) but, in the board's view, did not have an impact on the considerations applying to the issue. The question whether post-published evidence could be taken into account on substantive grounds, depending on the plausibility of the technical effect based on the evidence submitted as proof, was extensively discussed in **T 116/18**. This discussion touched on aspects relating to monopoly, the notion of technical contribution and the weight to be attached to speculations.

See also in this chapter II.C.7.2. "Level of disclosure required for medical use – plausibility".

7. The requirement of sufficiency of disclosure in the biotechnology field

7.1. Clarity and completeness of disclosure

7.1.1 General

The principles elucidated under chapter II.C.4. and 5. above are also applicable to biological inventions. In particular, reference should be made to the case law laid down by the boards in **T 281/86** (OJ 1989, 202), **T 299/86** of 17 August 1989 and **T 409/91** (OJ 1994, 653). Issues related to completeness of disclosure are also discussed by the boards in context with inventive step (see e.g. **T 1329/04**, **T 604/04**, **T 898/05**) and industrial applicability (see e.g. **T 870/04**, **T 641/05**, **T 1452/06**, above chapter I.E.). Whether the application discloses sufficient information making it plausible that the claimed polynucleotides or polypeptides have the alleged technical effect was considered a matter of inventive step (**T 743/97**; **T 1329/04**) or industrial applicability (**T 1165/06**,

T 1452/06), whereas the relevant question under Art. 83 EPC 1973 was whether the description was sufficiently clear and complete for the skilled person to prepare the claimed products (**T 743/97**).

In **T 449/90**, the board considered that the requirements of Art. 83 EPC 1973 had been satisfied where the claimed degree of inactivation ("substantially") of the Aids virus could be demonstrated with sufficient certainty. Complete inactivation of the life-threatening virus – which the opponent had argued was necessary – was indeed highly desirable, but not an issue under Art. 83 EPC 1973, given the claim as worded.

7.1.2 One way of implementing invention over whole scope of claim

When examining sufficiency of disclosure, the boards have to be satisfied, firstly, that the patent specification places the skilled person in possession of **at least one way** of putting the claimed invention into practice, and secondly, that the skilled person can put the invention into practice **over the whole scope of the claim** (see e.g. **T 792/00**, **T 811/01**, **T 1241/03**, **T 364/06**; see also **T 1727/12** on the notion of "Biogen sufficiency" and **T 1845/14**, in which the board confirmed that, as indicated in **T 1727/12**, "so-called Biogen insufficiency" is not part of the established case law of the boards of appeal). The scope of the patent should be justified by the technical contribution to the art (**T 612/92**). The necessary extent of disclosure is assessed on a case-by-case basis having regard to the essence of the invention (**T 694/92**, OJ 1997, 408).

In **T 292/85** (OJ 1989, 275) the board stated that an invention is regarded as sufficiently disclosed if at least one way is clearly indicated enabling the skilled person to carry out the invention. The invention at issue concerned a recombinant plasmid comprising a homologous regulon, heterologous DNA and one or more termination codons for expression in bacteria of a functional heterologous polypeptide in recoverable form. The application was refused by the examining division on the grounds that not all embodiments falling within the broad functional wording of the claims were available. The board, however, held that the non-availability of some particular variants was immaterial as long as there were suitable variants known which provided the same effect.

Similarly, in **T 386/94** (OJ 1996, 658) the patent specification provided a technically detailed example for the expression of preprochymosin and its maturation forms in E. coli. It suggested the possibility of expressing these proteins in micro-organisms in general. The board held that the invention was sufficiently disclosed because one way to carry out the invention was clearly indicated and the state of the art contained no evidence that foreign genes could not be expressed in organisms other than E. coli. The principles set out in **T 292/85** (OJ 1989, 275) were also applied in **T 984/00** (where the invention lay in the use of the T-region of the Agrobacterium without the genes of the T-region of wild type Ti-plasmids to avoid the deleterious effects of these genes on the target plant) and in **T 309/06** (where the appellant had disclosed a novel group of enzymes characterised by useful properties and the board allowed the appellant to claim the enzymes independently of their origin).

As for the amount of detail needed for a sufficient disclosure, this depends on the correlation of the facts of the case to certain general parameters, such as the character of the technical field and the average amount of effort necessary to put into practice a certain written disclosure in that technical field, the time when the disclosure was presented to the public and the corresponding common general knowledge, and the amount of reliable technical details disclosed in a document (see **T 158/91**; **T 694/92**, OJ 1997, 408; **T 639/95**; **T 36/00**; **T 1466/05**; **T 2220/14**).

7.1.3 Repeatability

An invention may also be sufficiently disclosed where results are **not exactly repeatable**. Variations in construction within a class of genetic precursors, such as recombinant DNA molecules claimed by a combination of structural limitations and functional tests, were immaterial to the sufficiency of disclosure provided the skilled person could reliably obtain some members of the class without necessarily knowing in advance which member would thereby be made available (**T 301/87**, OJ 1990, 335).

The claimed subject-matter in **T 657/10** included an "elite event", i.e. a particular event resulting from a random method (for which the expectations always range from nil to high) and having at least one surprising, advantageous property. There was ample jurisprudence of the boards of appeal on "elite events". Although the specific random methods and resulting products with (normal) average properties might well be known in the prior art, the presence of a particular product with an unexpected advantageous property might justify the recognition of an inventive step. However, the disclosure has to enable a skilled person to obtain the particular product resulting from the "elite event" without the need to repeat the random method de novo, i.e. he must be able to obtain the particular product without having to rely on pure chance again. In the case before the board these requirements were not fulfilled.

7.1.4 Broad claims

The mere fact that a claim is broad is not in itself a ground for considering the application as not complying with the requirement for sufficient disclosure under Art. 83 EPC. Only if there are serious doubts, substantiated by verifiable facts, may an application be objected to for lack of sufficient disclosure (see **T 19/90**, OJ 1990, 476, point 3.3 of the Reasons). See also **T 612/92**, **T 309/06**, **T 617/07**, **T 351/01**, **T 21/05**, **T 1188/06**, **T 884/06** and **T 364/06**.

In some cases, more technical details and more than one example were found necessary in order to support claims of a broad scope, for example where the essence of the invention was the achievement of a given technical effect by known techniques in different areas of application and serious doubts existed as to whether this effect could readily be obtained for the whole range of applications claimed, more technical details and more than one example may be required (see **T 612/92**; **T 694/92**, OJ 1997, 408; **T 187/93** and **T 923/92**). In **T 694/92** incomplete guidance was given. The claimed subject-matter concerned a method for genetically modifying a plant cell. In fact, the board held that the experimental evidence and technical details in the description were not sufficient for the